

# Extending the Spatio-Temporal Skew- $t$ Process for Threshold Exceedance Prediction

AR(2) Temporal Priors and Multi-Resolution Spline Basis Functions

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# Outline

- ① Motivation
- ② Model
- ③ Simulation
- ④ Ozone data
- ⑤ Conclusions

# Problem background

- Monitoring networks: limited coverage  $\rightarrow$  statistical extrapolation, with quantified confidence
- Monitoring and compliance
- Target:  $\Pr(Y > L)$  at **unmonitored** sites — the upper tail, not the mean field

| Ozone NAAQS (ppb)                                   | 1997 | 2008 | 2015 (current) |
|-----------------------------------------------------|------|------|----------------|
| Annual 4th-highest daily max 8-hr avg., 3-year mean | 80   | 75   | 70             |

# Why standard spatial methods fall short

- ① **Margin** — Gaussian: light-tailed, symmetric; ozone maxima: **heavy-tailed, right-skewed**
- ② **Extremal Dependence** (Coles, Heffernan, and Tawn, 1999):

$$\chi(h) = \lim_{c \rightarrow c^*} \Pr[ Y(\mathbf{s} + h) > c \mid Y(\mathbf{s}) > c ], \quad \text{Gaussian: } \chi(h) = 0 \quad \forall h$$

heavy tails + asymmetry + asymptotic dependence  $\Rightarrow$  **skew- $t$  process**

# Why not max-stable or EVT threshold models

- Max-stable likelihood: infeasible at 1,089 stations
- 75 ppb  $\approx$  0.92 sample quantile — not the far tail in EVT sense
- Subasymptotic setting (Huser and Wadsworth, 2022): joint tail at high but *finite* levels
- Skew- $t$  = location-scale mixture of GPs  $\in$  this family

# Baseline STP: model and parameters

Morris et al. (2017):

$$Y_t(\mathbf{s}) = \underbrace{\mathbf{X}_t(\mathbf{s})^\top \boldsymbol{\beta}}_{\text{regression mean}} + \underbrace{\lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})|}_{\text{skewness term}} + \underbrace{\sigma_t(\mathbf{s}) \nu_t(\mathbf{s})}_{\text{spatial process}}$$

- $\lambda \in \mathcal{R}$ : skewness ·  $z_t(\mathbf{s}) \sim N(0, 1)$ ,  $|z_t|$  half-normal
- $\sigma_t^2(\mathbf{s}) \sim \text{IG}(a/2, b/2)$ : scale mixing  $\rightarrow$  heavy tails ( $a = \text{df}$ )
- $\nu_t(\mathbf{s})$ : standard GP, Matérn correlation ( $\varphi$  range,  $\nu$  smoothness,  $\eta$  spatial proportion)
- Marginalize  $|z_t|$ ,  $\sigma_t \Rightarrow$  **skew- $t$  margin** + asymptotic dependence

# Baseline STP: Morris's extensions

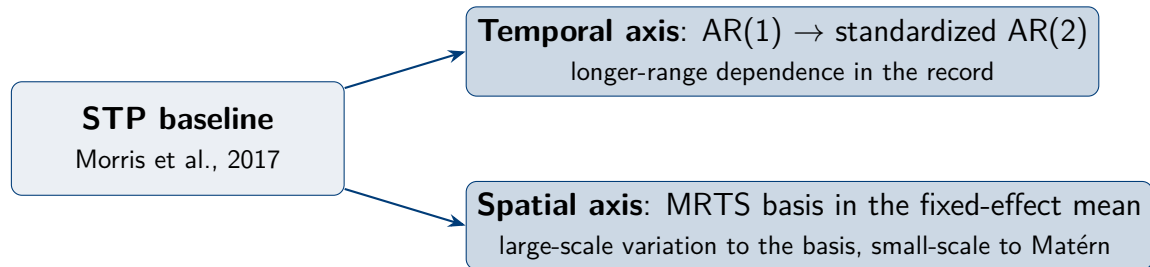
- ① **Random partition**,  $K$  knots redrawn daily:

$$P_{tk} = \{\mathbf{s} : k = \arg \min_{\ell} \|\mathbf{s} - \mathbf{w}_{t\ell}\|\}, \quad z_t(\mathbf{s}) = z_{tk}, \quad \sigma_t^2(\mathbf{s}) = \sigma_{tk}^2$$

→ **local** asymptotic dependence

- ② **Censored likelihood**:  $\tilde{Y}_t(\mathbf{s}) = \max\{Y_t(\mathbf{s}), T\}$ ,  $T \leq L$  — upper-tail focus; below  $T$ : imputed in MCMC
- ③ **Temporal dependence**: PIT → Gaussian scale → stationary **AR(1)**; margins preserved at every  $t$

# Two candidate extensions



candidate options, not replacements — held-out scores decide



# Extension 1: a standardized AR(2) prior

On the PIT-transformed Gaussian scale:

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \epsilon_t, \quad (\phi_1, \phi_2) \in \text{stationarity triangle}$$

Inverse PIT needs  $\gamma_0 = \text{Var}(Y_t) = 1 \Rightarrow \sigma_\epsilon^2$  **not free** (Yule–Walker):

$$\gamma_1 = \frac{\phi_1}{1 - \phi_2}, \quad \gamma_2 = \phi_1 \gamma_1 + \phi_2, \quad \sigma_\epsilon^2 = 1 - \phi_1 \gamma_1 - \phi_2 \gamma_2$$

- Initial pair  $(Y_1, Y_2)$ : stationary bivariate Gaussian, correlation  $\gamma_1$
- Identical margins across i.i.d. / AR(1) / AR(2)

## Extension 2: MRTS basis functions in the mean

MRTS (Tzeng and Huang, 2018):

- Ordered **coarse**  $\rightarrow$  **fine**:  $f_1 = 1$ ;  $f_2, f_3 =$  coordinates;  $k \geq 4$ : eigenvectors of the TPS kernel, eigenvalues decreasing
- Closed form at *new* locations  $\rightarrow$  predictive covariate

$$\mathbf{X}_t^*(\mathbf{s}) = [\mathbf{f}(\mathbf{s})^\top, \mathbf{X}_t(\mathbf{s})^\top]^\top, \quad Y_t(\mathbf{s}) = \mathbf{X}_t^*(\mathbf{s})^\top \boldsymbol{\beta}^* + \lambda \sigma_t(\mathbf{s}) |z_t(\mathbf{s})| + \sigma_t(\mathbf{s}) v_t(\mathbf{s})$$

- Claim: non-stationary **mean** only — Matérn keeps small-scale dependence

# Hierarchical model and inference

Three stages (Berliner, 1996):

**Stage 1** censored data:  $\tilde{Y}_t(\mathbf{s}_i) = \max\{Y_t(\mathbf{s}_i), T\}$

**Stage 2** Gaussian field · partition · latent AR(2) dynamics

**Stage 3** priors;  $(\phi_1, \phi_2) \sim \text{Unif}(\text{stationarity triangle})$

$$[\mathbf{Y}_{\text{cens}}, \mathcal{Z}, \boldsymbol{\theta} \mid \tilde{\mathbf{Y}}] \propto \prod_{t=1}^{n_t} [\tilde{\mathbf{Y}}_t \mid \mathbf{Y}_t] [\mathbf{Y}_t \mid \mathcal{Z}_t, \boldsymbol{\theta}] [\mathcal{Z}_{1:n_t} \mid \boldsymbol{\theta}] [\boldsymbol{\theta}]$$

- Gibbs-within-MH; AR(2) full conditionals: appendix
- $K$  fixed = **candidate model index**, chosen by held-out Brier

# Simulation design

144 sites on  $[0, 10]^2$  · 50 days · 10 data sets · 100 train / 44 test

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|                 |                                                                       |
|-----------------|-----------------------------------------------------------------------|
| Analysis models | Gaussian; skew- $t$ ( $K=1, 5$ ); sym- $t$ , $T=q(0.80)$ ( $K=1, 5$ ) |
|-----------------|-----------------------------------------------------------------------|

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|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| AR(2) designs | $(0.80, -0.35)$ , $(0.12, -0.05)$ at $K = 1, 5$ ;<br>persistent $(0.15, 0.80)$ ; ARFIMA $d = 0.45$ |
|---------------|----------------------------------------------------------------------------------------------------|

|              |                                                                               |
|--------------|-------------------------------------------------------------------------------|
| MRTS designs | 2× <b>second-moment</b> non-stationarity;<br>2× <b>nonlinear mean</b> surface |
|--------------|-------------------------------------------------------------------------------|

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# Evaluation protocols

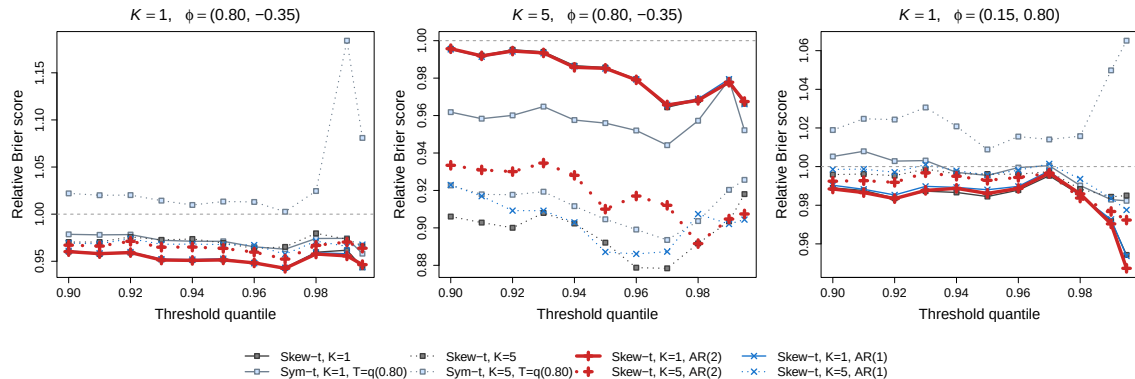
Scoring (Gneiting and Raftery, 2007):

$$\text{BS}(L) = \{I(y > L) - \hat{P}(Y > L)\}^2 \quad (\text{lower is better})$$

| Protocol            | Observed                    | Predicted                | Task          |
|---------------------|-----------------------------|--------------------------|---------------|
| Spatial hold-out    | training sites, full record | unseen sites             | interpolation |
| Time-block forecast | all sites, days 1–50        | same sites, next 15 days | extrapolation |

- Uncertainty: **paired** differences · 95% CI · Wilcoxon

# AR(2), interpolation: temporal structure barely moves the scores



32/33 paired intervals cover zero

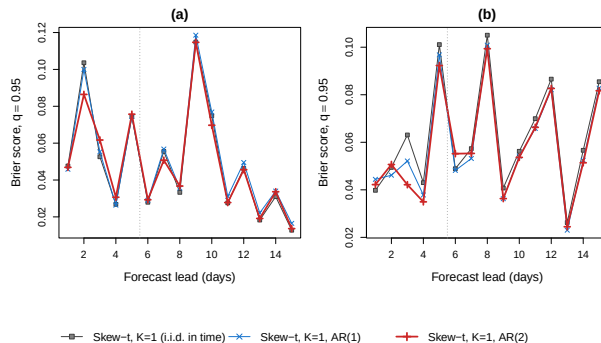
(all six settings: backup)

# AR(2): coefficients identified, scores still unmoved

| Generating design       | $\hat{\phi}^{(\sigma)}$ |                      | $\hat{\phi}^{(z)}$ |                      |
|-------------------------|-------------------------|----------------------|--------------------|----------------------|
|                         | $\hat{\phi}_1$          | $\hat{\phi}_2$       | $\hat{\phi}_1$     | $\hat{\phi}_2$       |
| Persistent (0.15, 0.80) | 0.112 (0.132)           | <b>0.742</b> (0.117) | 0.175 (0.152)      | <b>0.728</b> (0.112) |
| ARFIMA, $d = 0.45$      | 0.502 (0.237)           | <b>0.200</b> (0.239) | 0.365 (0.249)      | <b>0.156</b> (0.169) |

- Second lag **recovered** — even under long memory
- Hold-out scores: still unmoved  $\rightarrow$  the **task**, not the model

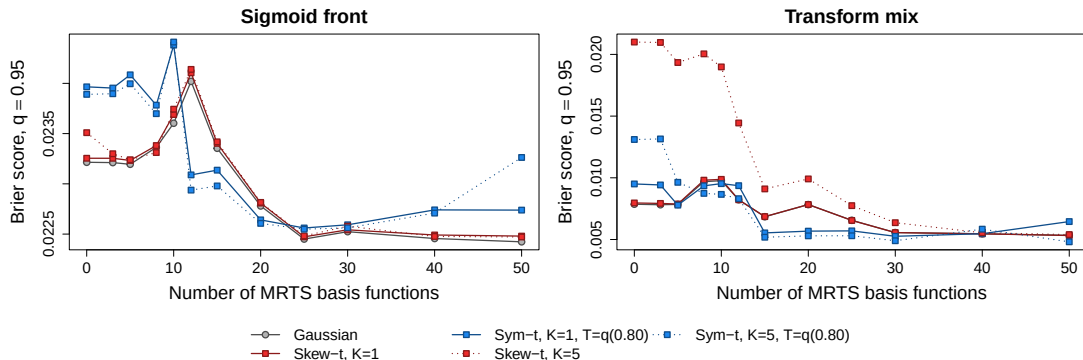
# AR(2), extrapolation: directional gains where predicted



- Long memory: AR(2) lowest at almost every lead; short-memory control: ties
- All differences n.s. → **directional** (largest  $-0.0068$ , CI  $[-0.033, 0.020]$ )



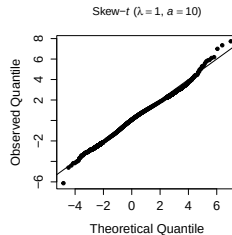
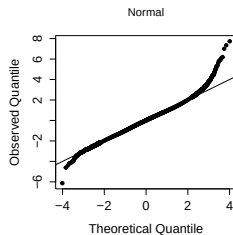
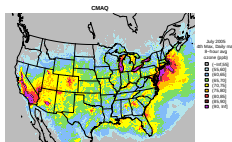
# MRTS: inert against second moments, paid off once resolved



- Second-moment designs: flat — **no first moment to recover** (backup)
- Nonlinear mean: pays only once the basis **resolves** the surface; mid-size dip

# Ozone data and motivating diagnostics

- July 2005 · AQS daily max 8-h ozone · **CMAQ** covariate
- Models **calibrate CMAQ** at the monitors



- Residuals: heavy-tailed, right-skewed
- Skew- $t$  reference ( $\lambda = 1$ ,  $a = 10$ ): straightens

# Candidate set and validation design

- Baseline grid (Morris et al., 2017):  $\{\text{Gaussian, skew-}t\} \times K \in \{1, \dots, 15\} \times T \in \{0, 50, 75\} \text{ ppb} \times \{\text{TS, no TS}\}$
- Extensions: AR(2) prior; MRTS with  $K_{\text{MRTS}} = 10$  **pre-specified**
- $1,089 \rightarrow 800$  **region-stratified** sites · two-fold CV (400/400) · splits 300/100, 200/200
- MCMC: 30,000 iterations, 25,000 burn-in

# Results: scattered wins, but systematic

| Level $L$ | Rank 1 model                                                        | Rank 2 model                            |
|-----------|---------------------------------------------------------------------|-----------------------------------------|
| 0.900     | <b>MRTS</b> skew- $t$ , $K=1$ , $T=0$ , AR(1), $K_{\text{MRTS}}=10$ | skew- $t$ , $K=1$ , $T=0$ , AR(1)       |
| 0.950     | <b>AR2</b> skew- $t$ , $K=7$ , $T=50$                               | <b>AR2</b> skew- $t$ , $K=10$ , $T=50$  |
| 0.980     | skew- $t$ , $K=6$ , $T=50$ , AR(1)                                  | skew- $t$ , $K=5$ , $T=50$ , AR(1)      |
| 0.990     | symmetric- $t$ , $K=9$ , $T=75$ , AR(1)                             | skew- $t$ , $K=5$ , $T=50$              |
| 0.995     | <b>AR2</b> symmetric- $t$ , $K=15$ , $T=75$                         | symmetric- $t$ , $K=9$ , $T=75$ , AR(1) |

- Deeper tail  $\rightarrow$  higher censoring, larger  $K$
- Pattern reproduces on the 300/100 and 200/200 splits

# Uncertainty: the two axes part ways

Site-clustered bootstrap ( $B = 2000$ , by fold), 400/400 split,  $\Delta \times 10^3$  [95% CI]:

| $L$ quantile | AR(2) – AR(1)        | MRTS – none                    |
|--------------|----------------------|--------------------------------|
| 0.900        | –0.15 [–0.38, +0.08] | –0.03 [–0.13, +0.07]           |
| 0.950        | –0.12 [–0.29, +0.04] | – <b>0.08</b> * [–0.14, –0.02] |
| 0.980        | +0.03 [–0.08, +0.14] | –0.04 [–0.07, +0.00]           |
| 0.990        | –0.06 [–0.14, +0.03] | – <b>0.03</b> * [–0.05, –0.00] |
| 0.995        | –0.03 [–0.08, +0.02] | –0.01 [–0.02, +0.01]           |

- AR(2): every interval covers zero (all splits)
- MRTS: negative everywhere; stars survive **block resampling**;  $\approx 3\text{--}4 \times 10^{-3}$  of baseline

# Conclusions

- **Temporal axis:** divides along the task — identified yet inert in interpolation; gains only in **forecasting**, never significant
- **Spatial axis:** **real, though modest** — needs mean-level structure, and a basis fine enough to resolve it
- **Contribution:** STP baseline → **candidate model class**, selected by held-out threshold prediction

# Limitations and future work

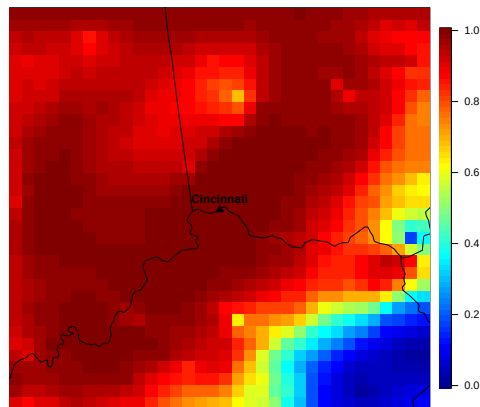
## Limitations

- One month, one pollutant · single forecast origin

## Future work

- **Random-coefficient** MRTS (fixed-rank kriging) → the second-moment layer
- Reparameterized  $z$  , longer records , rolling forecast origin

# An exceedance-probability map



Pr( $\geq 1$  day  $> 75$  ppb), July 2005, Cincinnati — skew- $t$  + TS + MRTS ( $K_{\text{MRTS}}=10$ )



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Huser, Raphaël and Jennifer L. Wadsworth (2022). “Advances in statistical modeling of spatial extremes”. en. In: *WIREs Computational Statistics* 14.1. \_eprint: <https://wires.onlinelibrary.wiley.com/doi/pdf/10.1002/wics.1537>, e1537. ISSN: 1939-0068. DOI: 10.1002/wics.1537. URL: <https://onlinelibrary.wiley.com/doi/abs/10.1002/wics.1537> (visited on 07/02/2026).

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Morris, Samuel A. et al. (Sept. 2017). “A Space-Time Skew-t Model for Threshold Exceedances”. In: *Biometrics* 73.3, pp. 749–758. ISSN: 0006-341X. DOI: 10.1111/biom.12644. URL: <https://doi.org/10.1111/biom.12644> (visited on 03/14/2025).



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# Thank You

Questions & Comments Welcome

Backup

# Backup: $\lambda$ - $z$ reflection ridge and forecast screening

- A reflection-type non-identifiability links  $\lambda$  and  $z$ : short-record fits can land on the **reflected ridge**, where the skewness parameter and the intercept fail to separate and the predictive spread inflates.
- The screening rule is public and stated with the forecast protocol: fits whose  $\lambda$  and intercept fail to separate, or whose predictive spread is unbounded, are removed *before* scoring.
- Root fix — a sign-anchored reparameterization of  $z$  — is listed as future work.

# Backup: AR(2) stationary moments and full conditionals

- Yule–Walker under  $\gamma_0 = 1$ :  $\gamma_h = \phi_1\gamma_{h-1} + \phi_2\gamma_{h-2}$  for  $h \geq 1$  gives  $\gamma_1 = \phi_1/(1 - \phi_2)$ ,  $\gamma_2 = \phi_1\gamma_1 + \phi_2$ , and  $\sigma_\epsilon^2 = 1 - \phi_1\gamma_1 - \phi_2\gamma_2$ .
- Gibbs-within-MH: the full conditional of  $Y_t$  carries **three** prior factors,

$$\pi(Y_t \mid Y_{-t}, \mathbf{Y}) \propto L(\mathbf{Y} \mid Y_t) \underbrace{p(Y_t \mid Y_{t-1}, Y_{t-2})}_{\text{self}} \underbrace{p(Y_{t+1} \mid Y_t, Y_{t-1})}_{\text{lag-1}} \underbrace{p(Y_{t+2} \mid Y_{t+1}, Y_t)}_{\text{lag-2}};$$

omitting the lag-2 factor distorts the stationary distribution.

- Head of the series ( $t = 1, 2$ ): the stationary initial-pair density replaces the self factor. Proposals: symmetric Gaussian random walk tuned to 0.3–0.4 acceptance;  $(\phi_1, \phi_2)$  updated jointly under the uniform prior on the stationarity triangle.

# Backup: second-moment design — “inert” is not “no signal”

- The random effect superposes two fixed cosine fields with daily-redrawn Gaussian weights:

$$u_t(\mathbf{s}) = \sum_{j=1}^2 \xi_{tj} g_j(\mathbf{s}), \quad \xi_{tj} \stackrel{iid}{\sim} N(0, \tau_j^2), \quad (\tau_1, \tau_2) = (5, 3),$$

giving the location-dependent covariance  $\text{Cov}\{u_t(\mathbf{s}), u_t(\mathbf{s}')\} = \sum_j \tau_j^2 g_j(\mathbf{s}) g_j(\mathbf{s}')$ .

- Zero time-average  $\Rightarrow$  the pooled coefficient has **no first-moment structure to recover**, while  $u_t$  still contributes substantial non-stationary variance: structurally out of reach for a fixed-effect mean, not absent signal.
- The two nonlinear-mean surfaces were confirmed by an **oracle calculation before any fitting** to leave genuine Brier headroom.

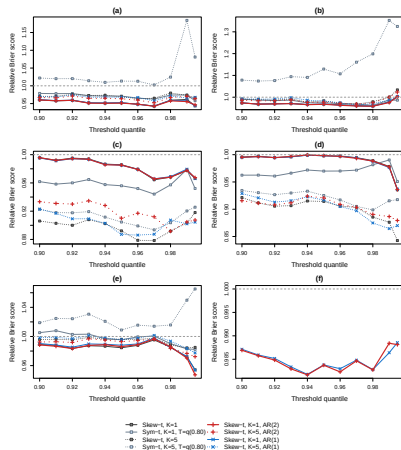


# Backup: censoring pairing and MCMC settings

- Why symmetric- $t$  with censoring: once the lower tail is censored, the skewness parameter is only **weakly identified**, so censored settings pair with a symmetric- $t$  marginal, following Morris et al. (2017).
- Thresholds: the censor level  $T = 75$  ppb sits below the era standard  $L = 80$  ppb ( $T \leq L$ ), and every evaluation level lies above  $T$ ; the degrees of freedom  $a$  control the tail thickness.
- Burn-in 25,000 of 30,000 ( $\approx 83\%$ ): mixing is slowed by censoring imputation and the latent volume; traces were inspected, and the long burn-in is a conservative choice.

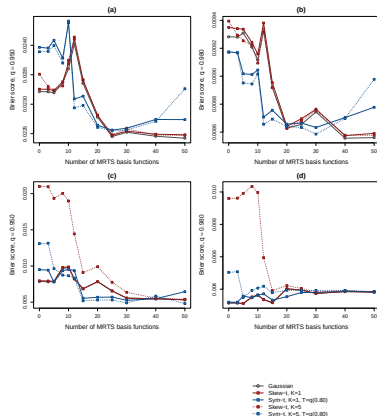
# Backup: spatial hold-out, all six generating settings

Relative Brier score vs. the Gaussian reference ( $\equiv 1$ , dashed).



- (a)–(d): fixed AR(2) designs,  $K \in \{1, 5\}$ ,  $(\phi_1, \phi_2) = (0.80, -0.35)$  or  $(0.12, -0.05)$ .
- (e): persistent (0.15, 0.80).
- (f): ARFIMA,  $d = 0.45$  (only the  $K=1$  AR(1)/AR(2) fits).

# Backup: MRTS nonlinear-mean designs, full grid



Brier score vs.  $K_{\text{MRTS}}$  ( $K_{\text{MRTS}} = 0$  is the no-MRTS baseline).

- (a, b): sigmoid front,  $q = 0.95$  / 0.98.
- (c, d): transform mix,  $q = 0.95$  / 0.98.
- At  $q = 0.98$  the gain is carried by the five-knot models — the sharpness–extrapolation signature.